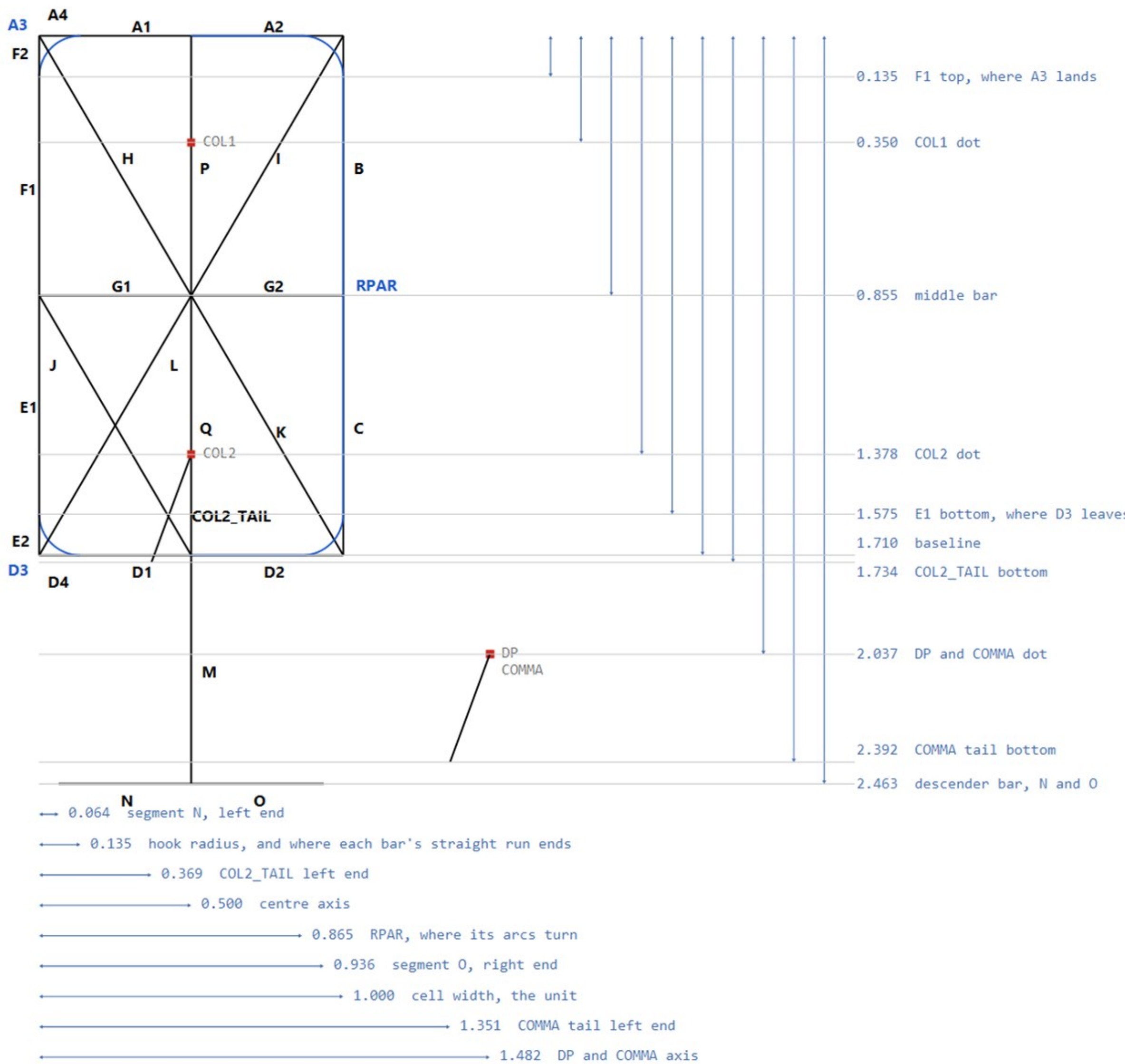
TalkRPN cell - centreline skeleton

Inspired by the HP-01 (1977) and the Litronix DL-3422, and identical to neither. The HP-01's look - hooked corners, the slant, the proportions - carried onto roughly the DL-3422's segment count, so the display can show letters as well as digits. Departures from both are deliberate: the corners hook where the DL-3422's are square, the segments meet flush where a real part leaves gaps, the decimal point sits between cells rather than taking one of its own, and there are elements here that neither part had. It is a period style, not a reproduction of any real component.

THE UNIT: segment E/F to segment B/C is 1 - the left column to the right column, centre to centre. Every length here is in that unit, across and down alike, so the centre axis is exactly 0.5 and pitch minus 1 is the clearance between neighbouring cells.

No stroke width, no slant: these are centrelines, and both are applied at render time. Origin is the top-left centreline corner; x right, y down.
28 bars + COL1 + COL2 + DP + COMMA = 32 elements; the mask is a Long. Blue: A3/D3 are ALTERNATIVES to A4/D4; RPAR is the whole right paren.
TalkRpnFont.kt is the source of truth - this sheet mirrors it.



TalkRPN cell - centreline listing

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SEGMENT CENTRELINES              from                  to
  A1    line         0.500,  0.000  ->     0.135,  0.000
  A2    line         0.500,  0.000  ->     1.000,  0.000
  A4    line         0.135,  0.000  ->     0.000,  0.000
  A3    arc          0.135,  0.000  ->     0.000,  0.135   arc R0.135 centre 0.135,0.135
  B     line         1.000,  0.000  ->     1.000,  0.855
  C     line         1.000,  0.855  ->     1.000,  1.710
  D1    line         0.500,  1.710  ->     0.135,  1.710
  D2    line         0.500,  1.710  ->     1.000,  1.710
  D4    line         0.135,  1.710  ->     0.000,  1.710
  D3    arc          0.135,  1.710  ->     0.000,  1.575   arc R0.135 centre 0.135,1.575
  F2    line         0.000,  0.000  ->     0.000,  0.135
  F1    line         0.000,  0.135  ->     0.000,  0.855
  E1    line         0.000,  0.855  ->     0.000,  1.575
  E2    line         0.000,  1.575  ->     0.000,  1.710
  G1    line         0.000,  0.855  ->     0.500,  0.855
  G2    line         0.500,  0.855  ->     1.000,  0.855
  H     line         0.000,  0.000  ->     0.500,  0.855
  I     line         1.000,  0.000  ->     0.500,  0.855
  J     line         0.000,  0.855  ->     0.500,  1.710
  L     line         0.500,  0.855  ->     0.000,  1.710
  K     line         0.500,  0.855  ->     1.000,  1.710
  P     line         0.500,  0.000  ->     0.500,  0.855
  Q     line         0.500,  0.855  ->     0.500,  1.710
  COL2_TAIL line         0.500,   1.378   ->     0.369,   1.734
  M     line         0.500,  1.710  ->     0.500,  2.463
  N     line         0.064,  2.463  ->     0.500,  2.463
  O     line         0.500,  2.463  ->     0.936,  2.463
  RPAR  compound     0.500,0.000 -> arc -> 1.000, 0.135..1.575 -> arc -> 0.500,1.710   the whole right paren

DOT CENTRES     COL1    0.500, 0.350      upper colon dot; also dots i and j
                COL2    0.500, 1.378      lower colon dot, colon only
                DP      1.482, 2.037     decimal point, outside the cell to the right
                COMMA   1.482, 2.037     same centre, plus a tail down and left
                SQUARE, side = 2 x STROKE = 0.295 - a real HP-55's decimal point is a square die

CENTRELINE BOX  1.000 wide x 1.710 tall    aspect 0.585
                2.463 tall including the descender
PITCH           2.430   origin to origin; minus 1 IS the clearance between cells
VPITCH          floor is 2.610, the ink height - below that descenders reach the next row
SLANT           7.5 degrees, applied at render

RELATIONSHIPS
  hook radius = hook start x = 0.135, so each arc is tangent to both bar and column
  F1 top = 0.135 = where A3 lands.  E1 bottom = 1.575 = where D3 lands.
  F2 and E2 are the stubs that carry the left column out to the corner when it is square.
  B meets C at y = 0.855: the right column is undivided at the corners, which is what makes a 4 lopsided.
  N and O are inset 0.0640 and 0.0641 from the columns - symmetric to within measurement.
  COL2_TAIL and the COMMA tail are the SAME tail, 0.355 down and 0.131 left, hung off two different dots.
  That is what makes a semicolon a colon whose lower dot grew a tail, at no cost in new geometry.

THE THREE LEFT-COLUMN ENDINGS
  square   A4 + F2      most letters
  hooked   A3 + F1      0 2 3 5 7 8 9, A C G O Q S, ( & and @   (F2 dark)
  short    F1 alone     digit 4, no A at all - left side sits 0.135 lower than the right
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TalkRPN character set

16 per row, code-point order 0x20-0x7E. Stroke 0.147, slant 7.5 deg. Red = lit, grey = unlit, cyan = cell bounds and baseline. Orange ? = low confidence: ' f t

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
2	sp	! P COL2	" F2 F1 P	# A4 A1 A2 G1 G2 N O P Q M	$ A3 A1 A2 F1 G1 G2 C D4 D1 D2 P Q M	% A3 A1 F1 G1 P I L G2 C D2 Q	& A4 A1 P H G1 E1 D3 D1 D2 C K	' ? I	(A1 A3 F1 E1 D3 D1	) RPAR	* H I L K P Q G1 G2	+ G1 G2 P Q	, COMMA	- G1 G2	. DP	/ I L
3	0 A3 A1 A2 F1 B E1 C D3 D1 D2	1 B C	2 A3 A1 A2 B G1 G2 E1 D3 D1 D2	3 A3 A1 A2 B G1 G2 C D3 D1 D2	4 F1 G1 G2 B C	5 A3 A1 A2 F1 G1 G2 C D3 D1 D2	6 A4 A1 A2 F2 F1 G1 G2 E1 E2 C D4 D1 D2	7 A3 A1 A2 B C	8 A3 A1 A2 F1 B G1 G2 E1 C D3 D1 D2	9 A3 A1 A2 F1 B G1 G2 C D3 D1 D2	: COL1 COL2	; COL1 COL2 COL2_TAIL	< I K	= G1 D4 D1	> H L	? A4 A1 A2 B G2 Q
4	@ A3 A1 A2 B C D3 D1 D2 E1 G1 Q	A A3 A1 A2 F1 B G1 G2 E1 E2 C	B A4 A1 A2 B C D4 D1 D2 G2 P Q	C A3 A1 A2 F1 E1 D3 D1 D2	D A4 A1 A2 B C D4 D1 D2 P Q	E A4 A1 A2 F2 F1 G1 E1 E2 D4 D1 D2	F A4 A1 A2 F2 F1 G1 E1 E2	G A3 A1 A2 F1 E1 D3 D1 D2 C G2	H F2 F1 B G1 G2 E1 E2 C	I A4 A1 A2 P Q D4 D1 D2	J B C E1 D3 D1 D2	K F2 F1 E1 E2 G1 I K	L F2 F1 E1 E2 D4 D1 D2	M F2 F1 H I B E1 E2 C	N F2 F1 H B K E1 E2 C	O A3 A1 A2 F1 B E1 C D3 D1 D2
5	P A4 A1 A2 F2 F1 B G1 G2 E1 E2	Q A3 A1 A2 F1 B E1 C D3 D1 D2 K	R A4 A1 A2 F2 F1 B G1 G2 E1 E2 K	S A3 A1 A2 F1 G1 G2 C D4 D1 D2	T A4 A1 A2 P Q	U F2 F1 B E1 C D3 D1 D2	V F2 F1 E1 E2 I L	W F2 F1 B E1 E2 C L K	X H I L K	Y H I Q	Z A4 A1 A2 I L D4 D1 D2	[A2 P Q M O	\ H K	] A1 P Q M N	^ L K	_ D4 D1 D2
6	` H	a G1 Q D4 D1 L	b F2 F1 E1 G1 D3 D1 Q	c E1 G1 D3 D1	d E1 G1 D3 D1 P Q	e E1 E2 G1 D4 D1 L	f ? A2 P Q G1 G2	g E1 G1 D3 D1 Q M N	h F2 F1 E1 E2 G1 Q	i Q COL1	j Q M N COL1	k F2 F1 E1 E2 G1 J	l P Q	m G1 G2 E1 E2 C Q	n E1 E2 J Q	o E1 G1 D3 D1 Q
7	p C G2 D2 Q M	q E1 G1 D3 D1 Q M O	r E1 E2 G1	s G1 J D4 D1	t ? P Q G1 G2 D2	u E1 D3 D1 Q	v E1 E2 L	w E1 E2 C L K	x J L	y E1 E2 D4 D1 Q M N	z G1 L D4 D1	{ A2 P G1 Q M O	| P Q M	} A1 P G2 Q M N	~ F1 A3 A1 P G2 B	